

2.5.24

EE25BTECH11058 - Mohana Krishna Sushma

Question:

If $|\mathbf{a}| = 2$, $|\mathbf{b}| = 7$ and $\mathbf{a} \times \mathbf{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

Solution:

Let the vector $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We are given:

$$\|\mathbf{a}\| = 2 \quad (1)$$

$$\|\mathbf{b}\| = 7 \quad (2)$$

$$\mathbf{c} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (3)$$

The squared magnitude of a vector $\mathbf{v}$ can be found using the matrix product $\mathbf{v}^\top \mathbf{v}$. We calculate the squared magnitude of $\mathbf{c}$:

$$\|\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{c} \quad (4)$$

$$= \begin{pmatrix} 3 & 2 & 6 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} \quad (5)$$

$$= (3)(3) + (2)(2) + (6)(6) \quad (6)$$

$$= 9 + 4 + 36 \quad (7)$$

$$= 49 \quad (8)$$

Therefore, the magnitude of $\mathbf{c}$ is:

$$\|\mathbf{c}\| = \sqrt{49} = 7 \quad (9)$$

The magnitude of the cross product is also defined by the relation:

$$\|\mathbf{c}\| = \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \quad (10)$$

where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Substituting the known values:

$$7 = (2)(7) \sin(\theta) \quad (11)$$

$$7 = 14 \sin(\theta) \quad (12)$$

Solving for $\sin(\theta)$:

$$\sin(\theta) = \frac{7}{14} = \frac{1}{2} \quad (13)$$

This implies that the angle θ is:

$$\theta = \frac{\pi}{6} \text{ or } 30^\circ \quad (14)$$

Angle Between a and b with Cross Product

